

Fall 2026 Exploratory Data Analysis with R

Lecture 2: Data Manipulation and the Tidyverse

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Recap & Agenda

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- 02 The Pipe & Core Verbs
- 03 Creating New Variables
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Quick Recap: Last Week

- Got Posit Cloud set up and wrote our first lines of R
- Imported a dataset with `read_csv()`
- Took a first look at `unemp_df`: the U.S. unemployment rate, 1948–2022
- Got a first taste of `summary()` and `filter()`

Today: turn that first taste into real fluency.

Goals for Today

- Meet the pipe, `%>%`
- Learn the five core **dplyr** verbs: `select()`, `filter()`, `arrange()`, `mutate()`, `summarize()`
- Pair `group_by()` with `summarize()` to answer questions *by group*
- Reshape data from wide to long with `pivot_longer()`
- Apply all of it to `unemp_df` — and to the population data from Homework 1

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The Pipe & Core Verbs

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Reading Code Left to Right

Nested function calls get hard to read fast:

```
arrange(filter(unemp_df, Year > 2000), desc(Unrate))
```

```
# A tibble: 22 × 2
  Year Unrate
<dbl> <dbl>
1  2010   9.61
2  2009   9.28
3  2011   8.93
4  2020   8.09
5  2012   8.07
6  2013   7.36
7  2014   6.16
8  2003   5.99
```

```
9 2008 5.8
10 2002 5.78
# i 12 more rows
```

To read this you have to work from the inside out. The **pipe operator**, `%>%`, lets us write the same thing as a sequence of steps, read top to bottom:

```
unemp_df %>%
  filter(Year > 2000) %>%
  arrange(desc(Unrate))
```

```
# A tibble: 22 × 2
  Year Unrate
<dbl> <dbl>
1 2010 9.61
2 2009 9.28
3 2011 8.93
```

```
4 2020 8.09
5 2012 8.07
6 2013 7.36
7 2014 6.16
8 2003 5.99
9 2008 5.8
10 2002 5.78
# i 12 more rows
```


How to Read `%>%`

- Read it as “**and then**”
- `x %>% f()` is exactly the same as `f(x)`
- `x %>% f() %>% g()` is exactly the same as `g(f(x))`

Every function in the **tidyverse** is built to work this way: take data in as the first argument, hand data back out, ready for the next step.

Choosing Columns: `select()`

Sometimes a dataset has more columns than we need. `select()` keeps only the ones we ask for.

```
unemp_df %>%  
  select(Year, Unrate) %>%  
  head(4)
```

```
# A tibble: 4 × 2  
  Year Unrate  
  <dbl> <dbl>  
1  1948   3.75  
2  1949   6.05  
3  1950   5.21  
4  1951   3.28
```

`unemp_df` only has two columns already, so this one won't look like much yet — but wait until we bring in a dataset with more of them.

`select()` Tricks

- `select(df, -Year)` — keep everything *except* `Year`
- `select(df, starts_with("Un"))` — keep columns whose names start with `"Un"`
- Column order in, column order out: `select()` also **reorders**

Exercise 1: Filtering, Properly

Last week we used `filter()` once. Let's push it further.

1. Filter `unemp_df` to years **after** 2007 (the leadup to the financial crisis)
2. Filter to years where `Unrate` was **above 6**
3. Combine both conditions into a single `filter()` call

Combining Conditions

```
unemp_df %>%  
  filter(Year > 2007 & Unrate > 6)
```

```
# A tibble: 7 × 2  
  Year Unrate  
  <dbl> <dbl>  
1  2009   9.28  
2  2010   9.61  
3  2011   8.93  
4  2012   8.07  
5  2013   7.36  
6  2014   6.16  
7  2020   8.09
```

- `&` means **and** — both conditions must hold

- `|` means **or** — either condition can hold
- `%in%` checks membership: `Year %in% c(2008, 2009, 2020)`

Putting Things in Order: `arrange()`

```
unemp_df %>%  
  arrange(Unrate) %>%  
  head(5)
```

```
# A tibble: 5 × 2  
  Year Unrate  
  <dbl> <dbl>  
1  1953   2.92  
2  1952   3.02  
3  1951   3.28  
4  1969   3.49  
5  1968   3.56
```

That's the five **lowest**-unemployment years on record in this dataset. To flip the order:


```
unemp_df %>%  
  arrange(desc(Unrate)) %>%  
  head(5)
```

```
# A tibble: 5 × 2  
  Year Unrate  
  <dbl> <dbl>  
1  1982   9.71  
2  2010   9.61  
3  1983   9.6  
4  2009   9.28  
5  2011   8.93
```

arrange() + filter() Together

The real power shows up once you start chaining verbs:

```
unemp_df %>%  
  filter(Year >= 2000) %>%  
  arrange(desc(Unrate))
```

```
# A tibble: 23 × 2  
  Year Unrate  
  <dbl> <dbl>  
1  2010   9.61  
2  2009   9.28  
3  2011   8.93  
4  2020   8.09  
5  2012   8.07  
6  2013   7.36
```

```
7  2014    6.16
8  2003    5.99
9  2008    5.8
10 2002    5.78
# i 13 more rows
```

This reads as: *start with `unemp_df`, keep years from 2000 on, then sort by unemployment, highest first.*

Breakout Session 1

Task: Using `unemp_df` and the pipe, answer:

1. What are the three years with the **lowest** unemployment since 1990?
2. Was unemployment ever above 9% before the year 2000? Which years?
3. Sort all years from 1980–1989 by unemployment, worst to best.

Hint: every one of these is a `filter()` and/or `arrange()`, chained with `%>%`.

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Creating New Variables

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Creating New Columns: `mutate()`

Every verb so far has worked with columns that already exist. `mutate()` **creates** new ones.

```
unemp_df %>%  
  mutate(decade = (Year %/% 10) * 10) %>%  
  head(4)
```

```
# A tibble: 4 × 3  
  Year Unrate decade  
  <dbl> <dbl> <dbl>  
1  1948   3.75  1940  
2  1949   6.05  1940  
3  1950   5.21  1950  
4  1951   3.28  1950
```

`%%` is integer division — `1987 %% 10 = 198`, times 10 gives `1980`. This is a common trick for bucketing years into decades.

mutate() with lag()

A very common use case: comparing a row to the row *before* it.

```
unemp_df %>%  
  mutate(change = Unrate - lag(Unrate)) %>%  
  filter(Year >= 2007 & Year <= 2010)
```

```
# A tibble: 4 × 3  
  Year Unrate  change  
  <dbl> <dbl>   <dbl>  
1  2007   4.62 0.00833  
2  2008   5.8  1.18  
3  2009   9.28 3.48  
4  2010   9.61 0.325
```


`lag(Unrate)` shifts the whole column down by one row, so `change` is the year-over-year move in the unemployment rate. Watch what happens around 2009.

Take Ten

Back at the top of the hour.

(Refill your coffee. Stretch. Do not open Canvas.)

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Summarizing by Group

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`group_by()` + `summarize()`

The single most useful pattern in the tidyverse: split your data into groups, then compute something for each group.

```
unemp_df %>%  
  mutate(decade = (Year %/% 10) * 10) %>%  
  group_by(decade) %>%  
  summarize(avg_unrate = mean(Unrate),  
            max_unrate = max(Unrate))
```

```
# A tibble: 9 × 3  
  decade avg_unrate max_unrate  
  <dbl>    <dbl>    <dbl>  
1  1940      4.9      6.05  
2  1950      4.51     6.84  
3  1960      4.78     6.69
```

4	1970	6.22	8.48
5	1980	7.27	9.71
6	1990	5.76	7.49
7	2000	5.54	9.28
8	2010	6.22	9.61
9	2020	5.71	8.09

`group_by()` doesn't change what the data looks like — it changes what happens **next**. `summarize()` then collapses each group down to one row.

Exercise 2: By Decade

Using the `decade` column from `mutate()`:

1. Find the decade with the **highest average** unemployment
2. Find the single **worst year** within that decade
3. Do it all in one piped chain — no saving intermediate objects

Breakout Session 2

Work with a partner.

Task:

1. Compute average unemployment for every decade from 1950 through 2019
2. Sort the result from highest to lowest average
3. Which decade would you *least* want to be graduating into? Why?

Stretch goal: instead of `mean()`, try `sd()` — which decade had the most *volatile* unemployment, not just the highest?

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Reshaping Data

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Wide vs. Long Data

Remember `decade_pop.csv` from Homework 1? Let's bring it back.

```
pop_df
```

```
# A tibble: 24 × 4
  Year   Black   White Source
<dbl> <dbl>   <dbl> <chr>
1  1790  757208  3172006 Census https://www.census.gov/content/dam/Census/libr...
2  1800 1002037  4306446 Census https://www.census.gov/content/dam/Census/libr...
3  1810 1377808  5862073 Census https://www.census.gov/content/dam/Census/libr...
4  1820 1771656  7866797 Census https://www.census.gov/content/dam/Census/libr...
5  1830 2328642 10532060 Census https://www.census.gov/content/dam/Census/libr...
6  1840 2873648 14189705 Census https://www.census.gov/content/dam/Census/libr...
7  1850 3638808 19553068 Census https://www.census.gov/content/dam/Census/libr...
8  1860 4441830 26922537 Census https://www.census.gov/content/dam/Census/libr...
```

```
9  1870 4880009 33589377 Census https://www.census.gov/content/dam/Census/libr...
10 1880 6580793 43402970 Census https://www.census.gov/content/dam/Census/libr...
# i 14 more rows
```

This is **wide** data: each *row* is a decade, and *Black* and *White* are separate *columns*. It's easy to read, but awkward for the tidyverse — `ggplot()` and `group_by()` both want one **variable per column**, and here, “population” is really one variable split across two columns.

What We Want Instead

Long data would look like this:

Year	race	population
1790	Black	757208
1790	White	3172006
1800	Black	1002037
...	...	...

One row per Year–race combination. This is what `pivot_longer()` is for.

`pivot_longer()`

```
pop_long <- pop_df %>%  
  pivot_longer(cols = c(Black, White),  
               names_to = "race",  
               values_to = "population")  
  
pop_long %>% head(6)
```

```
# A tibble: 6 × 4  
  Year Source                                race population  
  <dbl> <chr>                                <chr>         <dbl>  
1  1790 Census https://www.census.gov/content/dam/Census/libra... Black          757208  
2  1790 Census https://www.census.gov/content/dam/Census/libra... White         3172006  
3  1800 Census https://www.census.gov/content/dam/Census/libra... Black          1002037  
4  1800 Census https://www.census.gov/content/dam/Census/libra... White         4306446
```

```
5 1810 Census https://www.census.gov/content/dam/Census/libra... Black 1377808
6 1810 Census https://www.census.gov/content/dam/Census/libra... White 5862073
```

- `cols` — which columns to fold together
- `names_to` — the new column that holds the old column *names* ("Black", "White")
- `values_to` — the new column that holds the values that used to live in those columns

Now `group_by()` Actually Works

```
pop_long %>%  
  group_by(race) %>%  
  summarize(pop_1790 = min(population),  
            pop_2020 = max(population),  
            growth = max(population) - min(population))
```

```
# A tibble: 2 × 4  
  race  pop_1790  pop_2020  growth  
  <chr>    <dbl>    <dbl>    <dbl>  
1 Black    757208  41100000  40342792  
2 White   3172006 223553265 220381259
```

That's the whole point of reshaping: once “race” is a column instead of two separate columns, `group_by(race)` is one line of code instead of a headache.

Breakout Session 3

Using `pop_long` from the last section:

1. Compute each group's **share** of the total population in 1900, 1950, and 2000 (`filter()` + `mutate()`)
2. Which decade had the fastest growth for each group? (Hint: you'll want `arrange()` inside each group — look up `group_by() %>% arrange()`)
3. In your own words: why was reshaping to long format a **necessary** step before you could answer question 2?

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Agentic Coding

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What Is Agentic Coding?

So far, every line of R in this class has been typed by a person. **Agentic coding** means prompting an LLM — Claude, in these examples — in plain English, and letting it write the code for you.

This is quickly becoming a normal part of professional data work, including at the Fed. It doesn't replace the tidyverse verbs from today — it depends on them.



Prompting Claude for a FRED Pull

In Homework 1 you set up a FRED account, requested an API key, and used `fredr_set_key()` to connect to it. Assume that's already done. Here's a prompt you might give Claude:

I already have a FRED API key set with `fredr_set_key()`. Using the `fredr` and `dplyr` packages, write R code that pulls the monthly `UNRATE` series back to 1990, buckets the observations by decade, and reports the average unemployment rate for each decade, sorted from highest to lowest.

Claude might hand back something like this:

```
library(fredr)
library(tidyverse)
library(lubridate)

unrate <- fredr(
  series_id = "UNRATE",
  observation_start = as.Date("1990-01-01")
)

unrate %>%
  mutate(decade = (year(date) %/% 10) * 10) %>%
  group_by(decade) %>%
  summarize(avg_unrate = mean(value, na.rm = TRUE)) %>%
  arrange(desc(avg_unrate))
```

Notice the shape: `mutate()`, `group_by()`, `summarize()`, `arrange()`. Four verbs from today's lecture, chained together. The agent didn't invent a new way to answer the question — it used the same toolkit you're building.

Checking the Agent's Work

`fredr` needs a live connection and your personal key, so we can't run Claude's exact code here. Notice, too, that the column names are different from what we're used to (`date` and `value`, not `Year` and `Unrate`) — that alone is worth catching before you trust the output.

We can still check the *logic* against data we already have loaded:

```
unemp_df %>%  
  filter(Year >= 1990) %>%  
  mutate(decade = (Year %/% 10) * 10) %>%  
  group_by(decade) %>%  
  summarize(avg_unrate = mean(Unrate)) %>%  
  arrange(desc(avg_unrate))
```

```
# A tibble: 4 × 2
  decade avg_unrate
  <dbl>    <dbl>
1  2010      6.22
2  1990      5.76
3  2020      5.71
4  2000      5.54
```

Same question, same answer shape. The point isn't that the numbers match exactly — FRED revises data, and today's date keeps moving — it's that you can independently reproduce the claim with verbs you already own, instead of taking it on faith.

Exercise 3: Prompt, Then Verify

1. Ask an LLM for R code, using `fredr`, that finds which decade since 1990 had the most **volatile** unemployment rate — not the highest average, the highest standard deviation.
2. Run the code it gives you.
3. Using only `mutate()`, `group_by()`, and `summarize()` on `unemp_df`, reproduce the same calculation yourself.
4. If the two don't agree, figure out why before you trust either answer.

Ground Rules for Agentic Coding

- **Never paste a real key or password into a prompt.** Treat a chat window like any other place code you didn't write yet runs. Use a placeholder like `fredr_set_key("YOURKEYHERE")`, exactly as Homework 1 does.
- **Read the code before you run it.** A confident-sounding agent can still call a function that doesn't exist, or reach for the wrong column name.
- **Verify with tools you understand.** Today's `mutate()`, `group_by()`, and `summarize()` are enough to catch most mistakes.
- Agentic coding multiplies the verbs you already have. It isn't a substitute for learning them.

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Recap

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Recap

Today we added five verbs to the toolkit:

- `select()` — choose columns
- `filter()` — choose rows
- `arrange()` — reorder rows
- `mutate()` — create columns
- `group_by()` + `summarize()` — compute *by group*

And one reshaping tool: `pivot_longer()`, for turning wide data into tidy, analysis-ready long data.

Chained together with `%>%`, these five verbs cover the overwhelming majority of day-to-day data wrangling — in this class, and afterward.

Links & Further Reading

- [R for Data Science, Ch. 5: Data Transformation](#)
- [RStudio dplyr cheat sheet](#)
- [tidyr pivoting vignette](#)

Coming up: more summary statistics, and starting the project.